DELHI PUBLIC SCHOOL, NAGPUR

Term 1 Examination 2026

Science · Class 10 A

Roll No. ☐☐☐☐☐☐☐☐

Time Allowed: 3 hours **Maximum Marks: 80**

GENERAL INSTRUCTIONS

1. All questions are compulsory unless a choice is indicated.
2. Marks for each question are indicated against it.
3. Read each question carefully before answering.
4. Write your answers in the answer booklet provided.
5. Draw neat, labelled diagrams wherever necessary.

SECTION A — OBJECTIVE (20 MARKS)

All questions are compulsory. Each carries 1 mark.

1. Which of the following is a decomposition reaction? **[1]**

(a) $2H_2 + O_2 \rightarrow 2H_2O$ (b) $CaCO_3 \rightarrow CaO + CO_2$

(c) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ (d) $NaOH + HCl \rightarrow NaCl + H_2O$

2. A solution turns red litmus blue. Its pH is likely to be **[1]**

(a) 1 (b) 4

(c) 5 (d) 10

3. Which metal is liquid at room temperature? **[1]**

(a) Sodium (b) Mercury

(c) Gallium (d) Zinc

4. The number of covalent bonds in a molecule of methane is **[1]**

(a) 2 (b) 3

(c) 4 (d) 5

5. Which substance in the stomach kills bacteria that enter with food? **[1]**

(a) bile (b) hydrochloric acid

(c) mucus (d) pepsin

6. The gap between two neurons is called a **[1]**

(a) dendrite (b) synapse

(c) axon (d) impulse

7. When an iron nail is dipped in copper sulphate solution, the colour of the solution changes from **[1]**

(a) green to blue (b) blue to green

(c) colourless to blue (d) blue to colourless

8. Which of the following is used to reduce the effect of an acidic bee sting? **[1]**

(a) vinegar (b) baking soda

(c) lemon juice (d) salt water

9. Which non-metal is a good conductor of electricity? **[1]**

(a) Sulphur (b) Graphite

(c) Phosphorus (d) Iodine

10. The functional group in ethanoic acid is **[1]**

(a) –OH (b) –CHO

(c) –COOH (d) >C=O

11. In which part of a cell does the breakdown of pyruvate into carbon dioxide and water take place? **[1]**

(a) cytoplasm (b) chloroplast
(c) mitochondria (d) nucleus

12. Which part of the brain maintains posture and balance? **[1]**

(a) cerebrum (b) cerebellum
(c) medulla (d) hypothalamus

13. The reaction $2AgCl \rightarrow 2Ag + Cl_2$ in sunlight is used in **[1]**

(a) black and white photography (b) making antacids
(c) whitewashing (d) electroplating

14. The chemical formula of bleaching powder is **[1]**

(a) $CaCO_3$ (b) $CaOCl_2$
(c) $Ca(OH)_2$ (d) $CaSO_4$

15. Metals that are kept under kerosene are **[1]**

(a) iron and copper (b) sodium and potassium
(c) gold and silver (d) zinc and aluminium

16. Which of the following is an unsaturated hydrocarbon? **[1]**

(a) Ethane (b) Propane
(c) Ethene (d) Butane

17. Assertion (A): The left ventricle has thicker walls than the right ventricle. Reason (R): The left ventricle pumps blood to the whole body. **[1]**

(a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false. (d) A is false, but R is true.

18. Assertion (A): Insulin deficiency causes diabetes. Reason (R): Insulin regulates the blood sugar level. **[1]**

(a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false. (d) A is false, but R is true.

19. Assertion (A): Burning of magnesium ribbon in air is a chemical change. Reason (R): A new substance, magnesium oxide, is formed. **[1]**

(a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false. (d) A is false, but R is true.

20. Assertion (A): Acids should be added to water slowly and not water to acids. Reason (R): Dilution of a concentrated acid is highly exothermic. **[1]**

(a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false. (d) A is false, but R is true.

SECTION B — VERY SHORT ANSWER (12 MARKS)

Each question carries 2 marks.

21. Why is sodium kept immersed in kerosene? **[2]**

22. What are isomers? Give one example. **[2]**

23. Why is diffusion insufficient to meet the oxygen needs of multicellular organisms like humans? **[2]**

24. What is a reflex arc? **[2]**

25. Why should a magnesium ribbon be cleaned before burning it in air? **[2]**

26. Why does dry HCl gas not change the colour of dry blue litmus paper? **[2]**

SECTION C — SHORT ANSWER (21 MARKS)

Each question carries 3 marks.

27. Show the formation of sodium chloride by the transfer of electrons. **[3]**

28. What is esterification? Write the equation for the reaction of ethanol with ethanoic acid and give one use of esters. **[3]**

29. Describe how food is absorbed in the small intestine. How is the intestine adapted for it? **[3]**

30. Describe the structure of a neuron with the function of each part. **[3]**

31. What is a displacement reaction? Write the equation for the reaction of iron with copper sulphate and state one observation. **[3]**

32. What is neutralisation? Write a balanced equation and give one use of neutralisation in daily life. **[3]**

33. Differentiate between roasting and calcination, with one example of each. **[3]**

SECTION D — LONG ANSWER (15 MARKS)

Each question carries 5 marks.

34. Describe the properties of ethanol and ethanoic acid. Give one test to distinguish between them and one harmful effect of drinking alcohol. **[5]**

35. Describe the structure and function of a nephron, and state how the amount of urine produced is regulated. **[5]**

36. Describe the main parts of the human brain and the function of each. How is the brain protected? **[5]**

SECTION E — CASE-BASED QUESTIONS (12 MARKS)

Each question carries 4 marks.

37. A student heats lead nitrate powder in a boiling tube. Brown fumes are given off and a yellow residue is left. (a) Name the brown gas. (b) Write the balanced equation. (c) Name the type of reaction. (d) Name the residue. **[4]**

38. Rina tested four solutions A, B, C and D with universal indicator and got pH 4, 1, 11 and 7. (a) Which is neutral? (b) Which is strongly acidic? (c) Which is strongly alkaline? (d) Arrange them in increasing order of H^+ ion concentration. **[4]**

39. A student placed strips of zinc, iron, copper and aluminium in solutions of each other's salts. (a) Which metal displaces all the others? (b) Which displaces none? (c) Write the reaction of iron with copper sulphate. (d) Arrange the four in decreasing reactivity. **[4]**

END OF QUESTION PAPER